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Disease Control and Toxicity Outcomes Comparing Ruthenium and Iodine Eye Plaque Brachytherapy in the Treatment of Intraocular Melanoma

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Purpose/Objective(s): Both Iodine-125 (^{125}I) COMS and Ruthenium-106 (^{106}Ru) eye plaques have the potential to achieve excellent tumor control in patients treated for uveal melanoma. We therefore evaluated our single institutional experience in the management of ocular melanoma treated with ^{125}I and ^{106}Ru plaque brachytherapy.

Materials/Methods: The records of 95 patients with uveal melanoma treated with either ^{106}Ru (n = 40) or ^{125}I (n = 55) plaque brachytherapy between 2000 and 2008 at UT MD Anderson Cancer Center were retrospectively reviewed. Endpoints assessed included overall survival (OS), tumor control (locoregional control (LRC), progression-free survival (PFS), and enucleation-free survival (EFS)), and toxicity (cataract formation, glaucoma development, visual loss). Patients were subdivided into groups based on pre-implant and post-implant visual acuity in the affected eye (20/20 up to 20/40, 20/40 up to 20/200, and worse than 20/200). Clinically significant visual loss was defined as movement between visual acuity groups.

Results: For patients receiving ^{106}Ru, median age at implant was 59 years, and median follow-up was 67 months (range, 6-101 months). Median lesion height was 3.05 mm (range, 1.27-4.94 mm) and median apical dose was 90 Gy with a median dwell time of 89 hours. For patients receiving ^{125}I, median age at implant was 62 years with a median follow-up of 55 months (range, 0-127 months). Median lesion height was 4.3 mm (range, 1.9-11.7 mm) with a median apical dose of 85 Gy with a median dwell time of 121 hours. Actuarial 5 year rates of LRC, PFS and OS with ^{106}Ru were 97%, 94%, and 92%. For ^{125}I, these values were 82%, 64%, and 80%. By log-rank analysis, LRC and PFS were different between the two groups ($p = 0.034$; $p = 0.001$), however OS was not different ($p = 0.051$). In the subset of patients with tumor apex height <5 mm (32 ^{125}I and all 40 ^{106}Ru), LRC and PFS remained improved with ^{106}Ru therapy ($p = 0.017$; $p = 0.013$) with no difference in OS. There were no enucleations in the ^{106}Ru cohort (EFS = 100%) and 9 in the ^{125}I group (5 year EFS = 80%). With ^{106}Ru treatment, 38% of patients experienced significant visual loss, 5% were diagnosed with glaucoma, and 50% developed cataracts. With ^{125}I, these values were 58%, 7% and 51%. In neither case was cataract formation associated with apical dose or apical dose rate.

Conclusions: Both ^{125}I and ^{106}Ru eye plaque brachytherapy treatment result in encouraging local control and survival for patients with intraocular melanoma. Our data represent the largest reported U.S. cohort of patients treated with ^{106}Ru plaque brachytherapy for uveal melanoma, and in comparison to our institutional experience with ^{125}I, suggest that ^{106}Ru plaque brachytherapy results in improved LRC and PFS with fewer enucleations, and less toxicity, supporting the reintroduction of this technique in the United States.

Author Disclosure: V. Takiar: None. K. Voong: None. D.S. Gombos: None. F. Mourtada: None. L. Rechner: None. A.A. Lawyer: None. W.H. Morrison: None. A.S. Garden: None. B.M. Beadle: None.

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Outcomes and Patterns of Recurrence of Orbital Carcinomas Treated With IMRT

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Purpose/Objective(s): The purpose of this study is to assess the outcomes of patients with malignant carcinomas of the orbit treated with intensity-modulated radiation therapy (IMRT).

Materials/Methods: We reviewed the medical records of 43 patients with carcinomas involving the orbit treated with IMRT at MD Anderson Cancer Center between 2002 through 2011. Survival outcomes were calculated using the Kaplan-Meier method with factors affecting survival determined by the log-rank test.

Results: Squamous cell and adenoid cystic carcinomas were the most common histologies (30% each) followed by basal cell carcinoma (19%). Nineteen patients (44%) had stage 3-4 disease and 18 (42%) had recurrent disease. Forty patients received surgery followed by adjuvant radiation therapy with doses ranging from 60-70 Gy (median 60 Gy) at 2-2.2 Gy/fraction. Eleven (28%) had eye-sparing surgery and 29 (72%) underwent orbital exenteration. Perineural invasion (PNI) was identified in 33 patients (77%) and close/positive margins were seen in 18 (45%). Three patients received definitive radiation therapy (range, 66-70 Gy). Eleven patients (26%) were treated to the neck, including all node positive patients. The median follow-up was 28 months (range, 4-117 months). Nine patients (21%) had local recurrences. Six recurrences were in-field and 3 occurred outside the treatment field (2 adjacent dermal recurrences and 1 in an unirradiated cavernous sinus). Of the 9 local recurrences, 5 occurred in patients with an exenteration and 4 in patients treated with an intact eye (including 1 treated with definitive radiation). Two patients had recurrence in the neck. Five-year overall survival (OS) and disease-free survival (DFS) rates for the entire group were 59% and 55%, respectively. On univariate analysis, PNI, involvement of a named nerve, and definitive radiation therapy without surgery were associated with a worse OS and DFS. Tumor histology did not significantly impact OS but patients with basal cell histologies had improved DFS rates compared to those with squamous cell and adenoid cystic histologies. One patient developed cataract but otherwise there were no late toxicities observed for the contralateral eye. Four of 14 patients developed severe (grade $\geq$3) toxicity of the treated intact eye; 3 required enucleation. The remaining 10 patients who received at least 60 Gy in 30 fractions to the eye experienced minimal (grade 1) or no toxicity.

Conclusions: Excellent rates of disease control for orbital carcinomas are achievable with IMRT combined with surgery. Toxicity for the contralateral eye was minimal; however, potential improvement may be achievable for treating the intact eye with other modalities such as proton therapy. Improved local control rates were observed with the combination of orbital exenteration and radiation therapy but the absolute loss of one eye must be weighed against preservation of binocular vision.

Author Disclosure: R. Tao: None. J. Phan: None. C.D. Fuller: None. G.B. Gunn: None. B.M. Beadle: None. S.J. Frank: None. W.H. Morrison: None. B. Esmaeli: None. M.E. Kupferman: None. A.S. Garden: None.

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Survival Outcomes of Re-irradiation With Intensity Modulated Radiation Therapy (IMRT) for Recurrent Head and Neck Squamous Cell Carcinoma

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Purpose/Objective(s): Assess the outcomes of curative-intent salvage reirradiation with IMRT for recurrent or second primary head and neck cancer in patients with previously irradiated head and neck squamous cell carcinoma.

Materials/Methods: A retrospective review was conducted from April 2004 - June 2012 of a total 88 patients with recurrent squamous cell carcinoma of the head and neck who underwent reirradiation at our institution utilizing IMRT. Sites of recurrent disease included: Neck (n = 48), Oropharynx (n = 34), Oral Cavity (n = 15), Larynx (n = 13), Hypopharynx (n = 7), Nasopharynx (n = 3), Salivary Gland (n = 2), Nasal Cavity (n = 1), and Sinus (n = 1). Of patients being treated, 29.5% had second primary cancers, 53.4% had local recurrences, and 17% had neck only recurrences. A total of 68.2% had surgical resection, and 79.5% had concurrent chemotherapy with reirradiation.

Results: Median follow-up of survivors was 15.2 months with median overall survival of 14.9 months (95% CI, 10.7-26.1). Median failure-free survival, defined as time from start of reirradiation to recurrence or death from treatment, toxicity, or uncontrolled tumor, was 10.8 months (95% CI, 5.2-24.3). A Cox proportional hazards regression analysis was used to assess predictors of overall survival and found that time to reirradiation